

Nuclear Fuel Performance

NE 533: Spring 2026

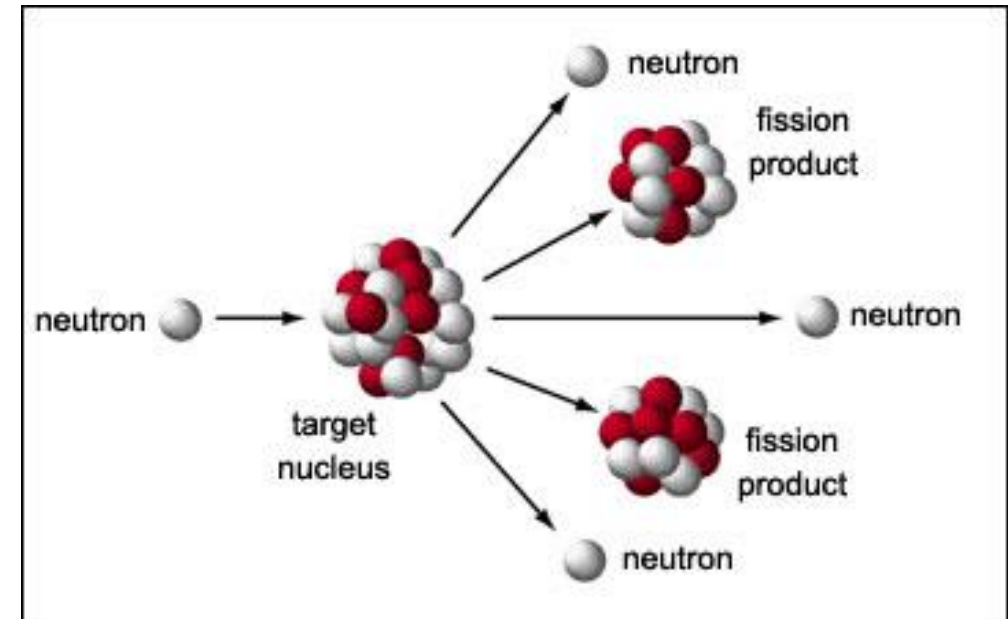
Last Time

- UO₂ has a number of good and bad characteristics
 - Good: high melting point, single phase, chemical compatibility, radiation resistant
 - Bad: brittle, low thermal conductivity, sensitive to stoichiometry, limited heating rates, etc.
- The “fuel system” consist of:
 - the fuel itself, the gap, the cladding, and the coolant
- Role of the cladding:
 - primary containment of FPs, chemical and physical barrier separating fuel from coolant
- Mining -> Processing -> Conversion -> Enrichment -> Powder -> Compaction/Sintering -> Rod/Assembly
- U₃O₈ must be converted to UF₆ for enrichment, which is then converted to UO₂ powder for pellet manufacture via sintering
- UO₂ microstructure from fabrication strongly impacts fuel performance

HEAT GENERATION

Fission basics

- Impinging neutron of a given energy
 - Neutron energy determines cross section which determines probability of fission event
- Neutron + Target Nucleus $\rightarrow$ Two fission products, 2-3 neutrons
- Fission releases around:
 - 210 MeV of energy
 - 170 MeV to fission fragments
 - 2 MeV per neutron
 - 7 MeV gamma rays
 - Balance radioactive decay



Energy release with different nuclei

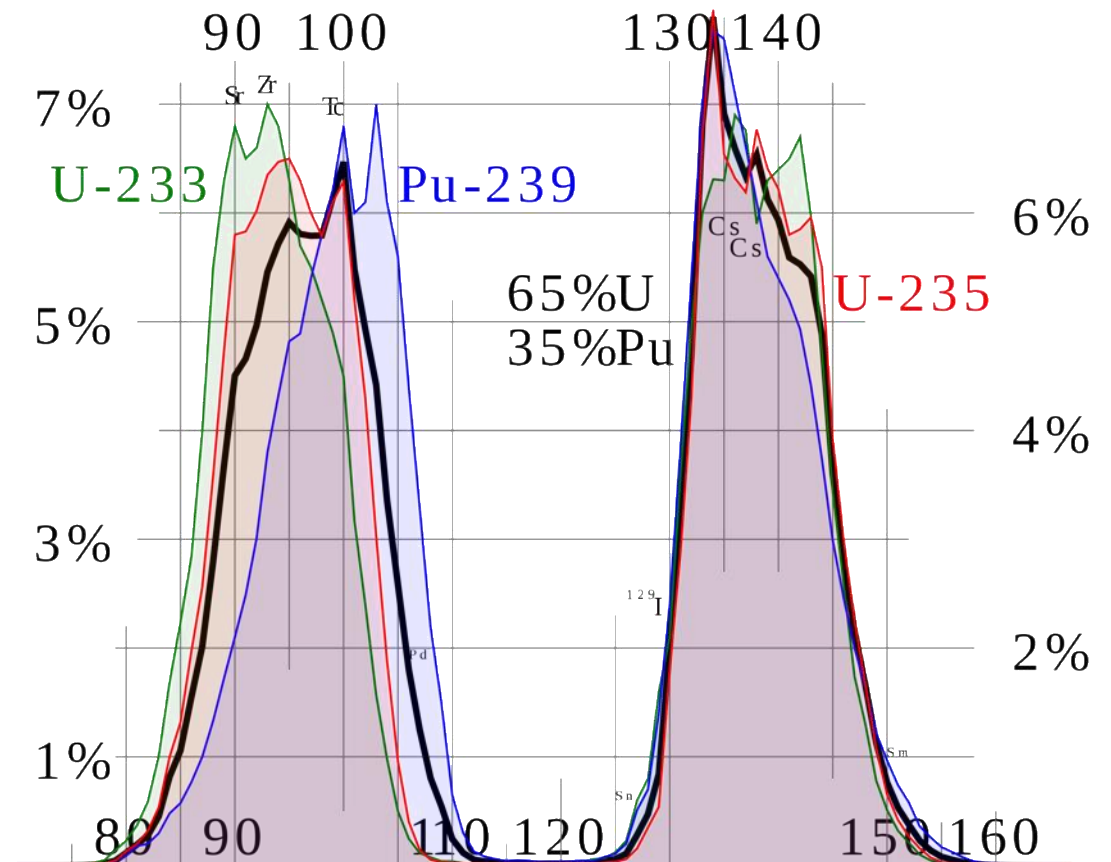
- Energy release is effectively agnostic with regards to the fissioning species
- Comparing U-235 with Pu-239 on the right
 - Pu releases about 9 MeV more usable energy per fission
 - Less than a 5% difference
- Partition of energy is largely identical as well

Source	Energy, MeV/f	
	²³⁵ U	²³⁹ Pu
Energy released instantaneously		
Kinetic energy of fission fragments	169.1	175.8
Kinetic energy of prompt neutrons	4.8	5.9
Energy of prompt γ -rays	7	7.8
Energy of γ -rays from $n\gamma$ capture	8.8	11.5
Energy from decay of fission products		
Energy of β^- -particles	6.5	5.3
Energy of delayed γ -rays	6.3	5.2
Energy of anti-neutrinos ¹	8.8	7.1
Total available energy	202.5	211.5

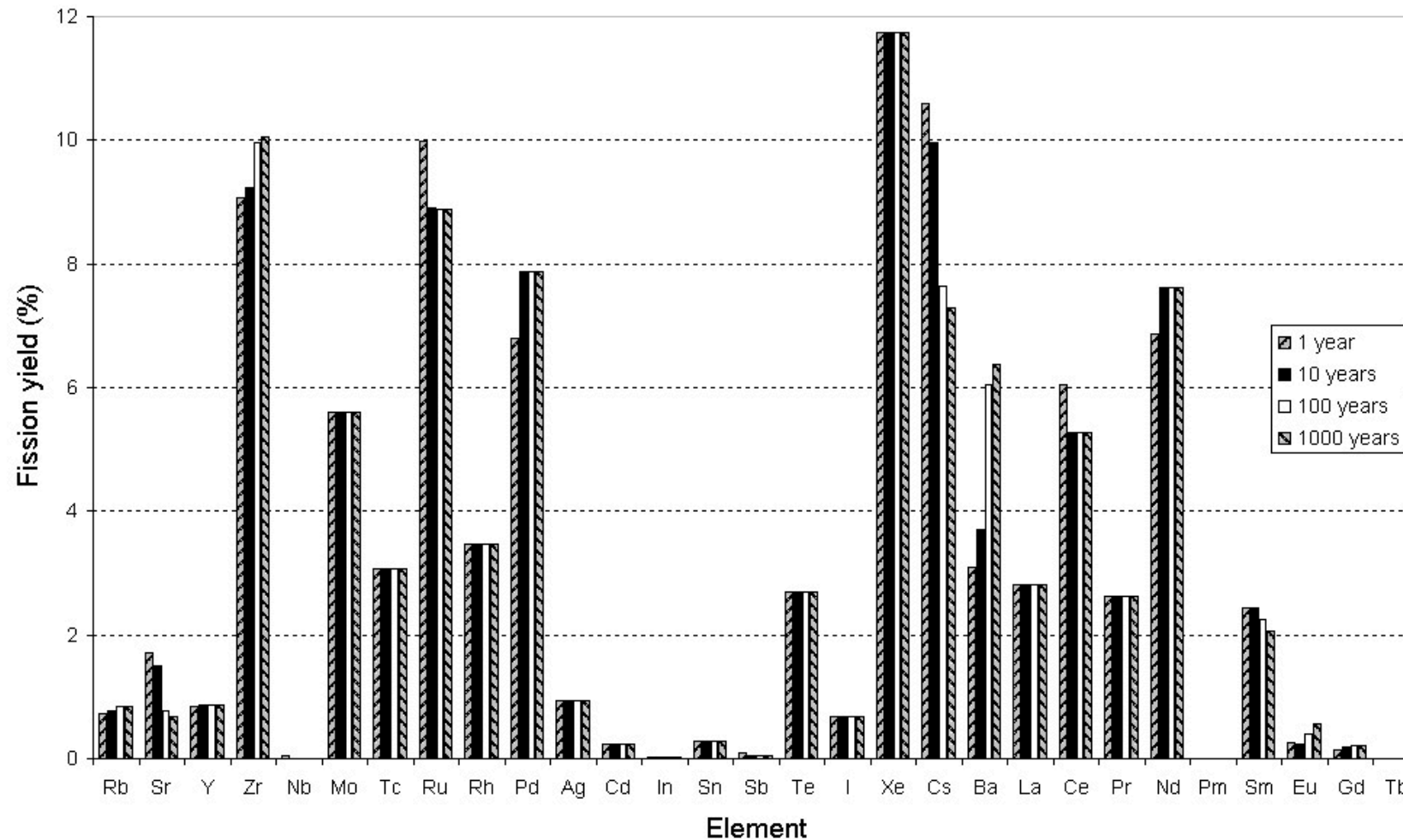
Note 1: Anti-neutrino energy is not absorbed in the reactor and does not contribute to the total available yield.

Fission product yield

- Regardless of fissioning isotope, fission product yields are effectively the same, in this double hump distribution
- One broad peak centered around $A=95$, the other around $A=135$
- Examples:
 - Mo ($Z=42$, $A=96$)
 - Cs ($Z=55$, $A=133$)



Fission Product Yields



Calculating heat generation rate for a given fuel

- We know about 200 MeV of energy is available due to a fission (210 MeV minus neutrinos)
- We know the fission cross section of the target nuclide (tabulated)
- We know the neutron flux (given by reactor conditions)
- We can calculate the fission atom density from the fuel type+enrichment
- The heat generation rate, Q is given by:
 - $Q = E_f \times N_f \times \sigma_f \times \phi$
 - Where E_f is the fission energy, N_f is the fission atom density, σ_f is the fission cross section, and ϕ is the neutron flux
 - Units: $\text{J/fission} \times \text{atoms/cm}^3 \times (\text{fission/neutron}) \times (\text{cm}^2/\text{atom}) \times (\text{neutron/cm}^2\text{-s}) = \text{J/cm}^3\text{-s} = \text{W/cm}^3$

Calculating heat generation rate for a given fuel

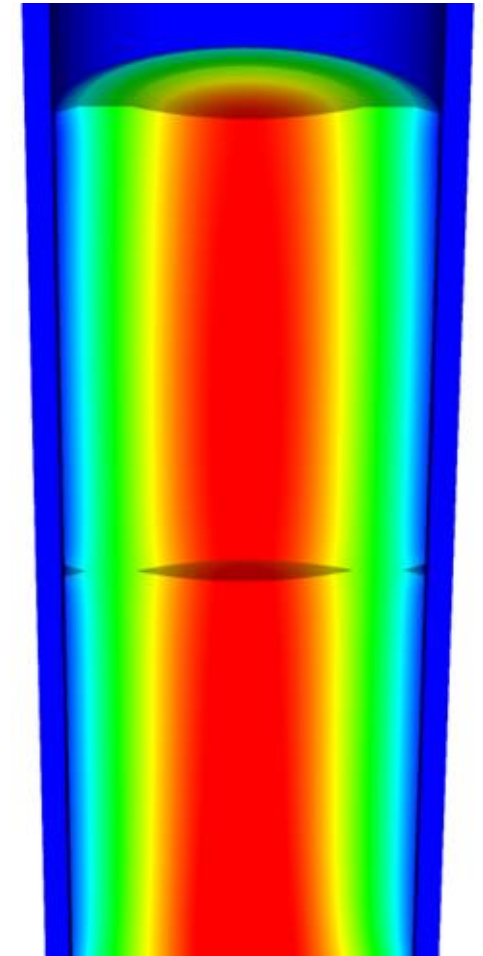
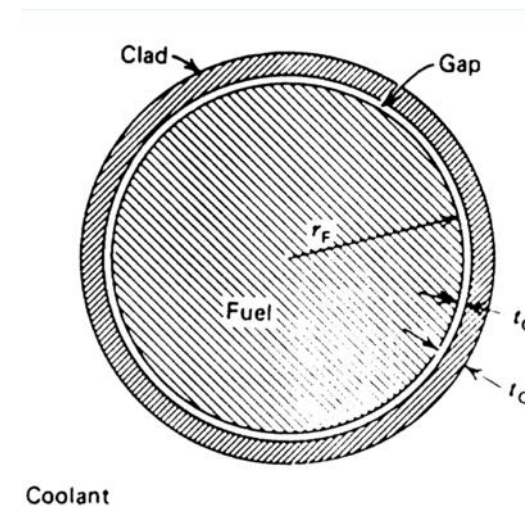
- Cross sections:
 - ENDF database: Nuclear Data Sheets 148 (2018) 1–142
 - Thermal neutron ($E=0.025$ eV) U235 fission cross section: ~ 586.8 barns
 - $1 \text{ barn} = 10^{-24} \text{ cm}^2$
- Fission atom density
 - Atom density of U-235 = UO₂ density x 1/molar mass x Avogadro's number x atom fraction x enrichment

Example

HEAT TRANSPORT

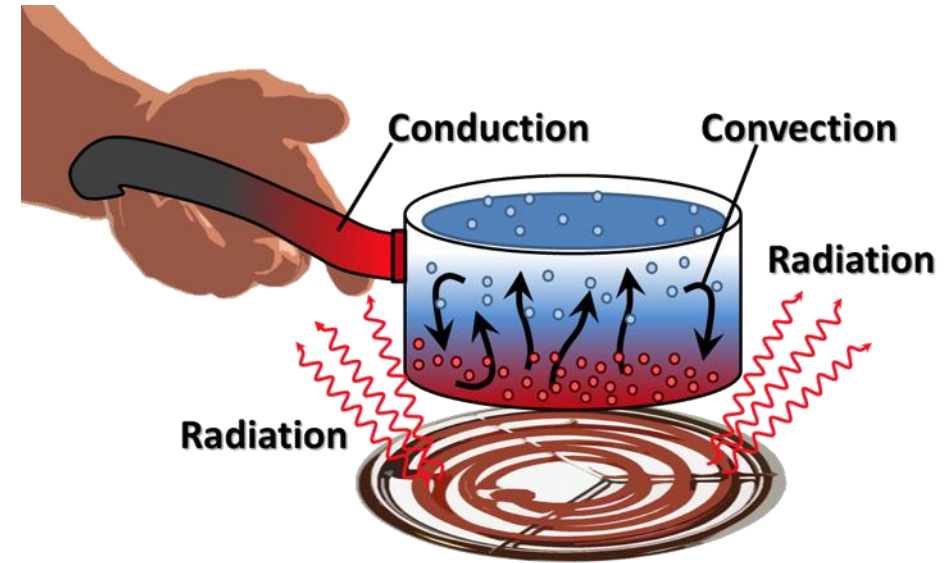
Heat transport route

- Heat is produced in the fuel, transports through the cladding and gap, and into the coolant
- Important quantities include
 - Volumetric heat generation rate Q (W/cm^3)
 - Fuel Centerline temperature T_0
 - Surface temperature of the fuel T_s
 - Inner cladding temperature T_{IC}
 - Outer cladding temperature T_{OC}
 - Coolant temperature T_{cool}
 - Fuel pellet radius r_F
 - Gap thickness t_G
 - Cladding thickness t_c
 - Coolant heat transfer h_c



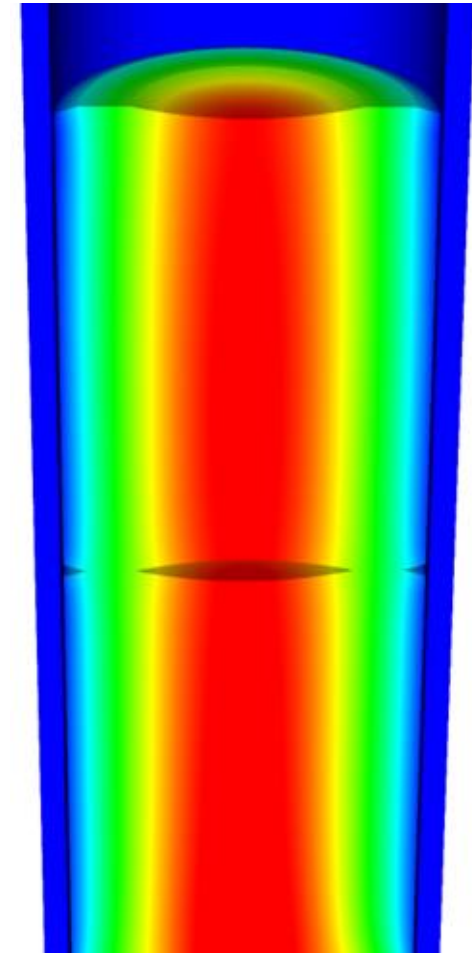
Heat can be transported in three ways

- Convection
 - Heat transfer through mass movement of liquid or gas
- Radiation
 - Heat transfer by means of photons in electromagnetic waves
- Conduction
 - Heat transfer by molecular, phonon, and electronic vibration/collisions



Heat transfer mode in fuel systems?

- How is heat transported through the fuel?
Conduction
- How is the heat transported through the gap?
Mostly conduction, some convection
- How is heat transported through the cladding?
Conduction
- How is heat transported to the coolant?
Convection



Heat conduction equation

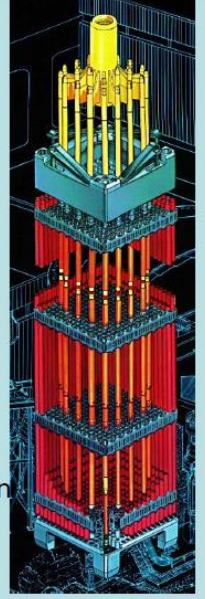
- ρ is the density, c_p is the specific heat, T is the temperature, t is the time, and k is the thermal conductivity, Q is heat generation
- It is a partial differential equation in time and space of the temperature, $T(\mathbf{x}, t)$, where $\mathbf{x}$ is a vector defining the position in space
- What do we need to know to solve this equation?

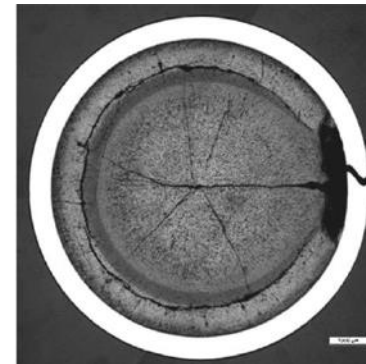
$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q$$

- The geometry of our problem
- The initial condition of T
- The boundary conditions of T
- Is each parameter a function of T
- If they aren't a function of T , do they vary in space and time for some other reason?

What is our geometry for the problem?

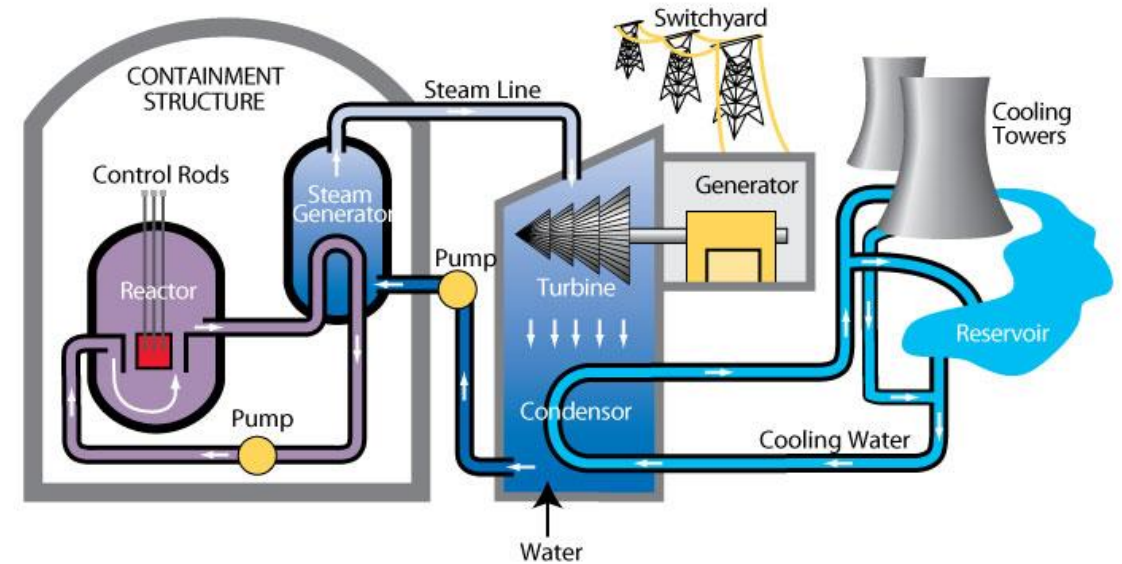
- Reactor geometry depends on reactor type
- The idealized geometry of each fuel rod is axisymmetric, but in reality, it is 3D
- Fuel pellet defects cause 3D geometry
- The stacked pellets may not be stacked perfectly, causing their center axis to not be aligned, also causing 3D geometry

	BWR	PWR	
	Lattice	10x10	
	Lattice size	~5.3"	
	Height	120"-150"	
	Fuel	UO ₂ /MOx	
	Fuel rods	~92	
	Part length rods	~14	
	Non-fueled rods	~2	
	Control	Ext. control rod	
	Cladding	Zr2	
	for PCI, nodular corrosion		
Channels	Yes	No	
Fuel mass	~180 kgU	~600 kgU	



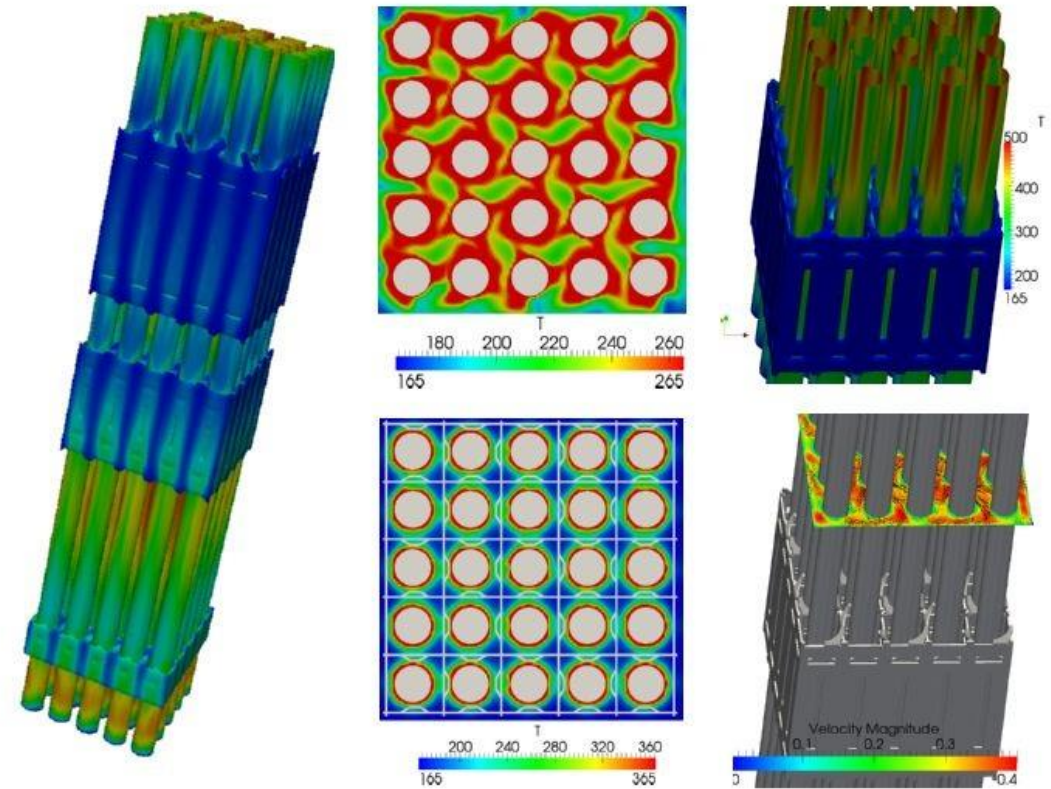
The initial condition of T

- The initial condition of T is set by the state of the reactor directly before startup (or before time of interest)
- The initial temperature is uniform throughout the fuel, equal to the initial coolant temperature
- $T(\mathbf{x}, 0) = T_{\text{cool}}(0)$



Boundary conditions?

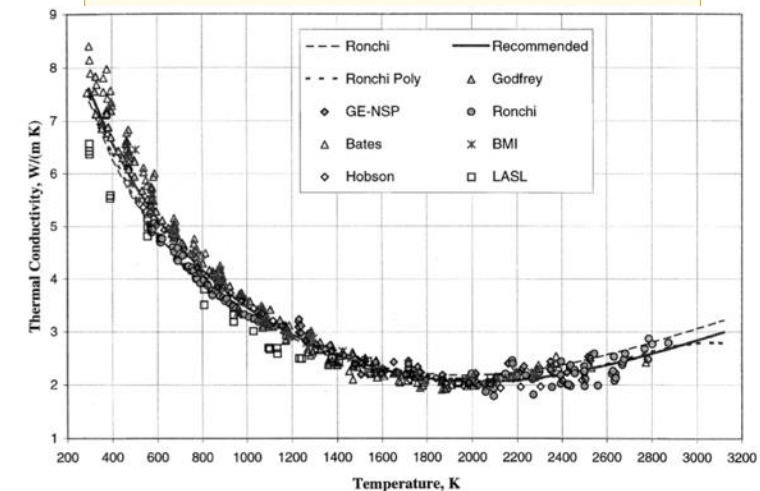
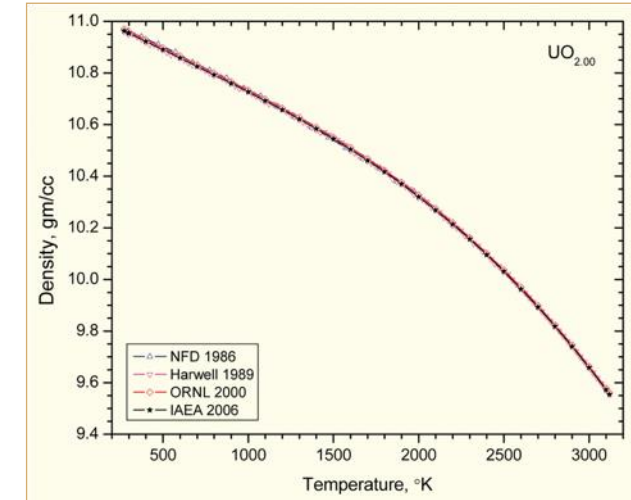
- The boundary conditions on T is set by the coolant flow
- The temperature of the coolant T_{cool} is complicated
 - It varies along the length of the fuel rod (axially)
 - It varies around the circumference of the fuel rod



Fuel properties

- All properties vary as a function of composition, thus as a function of burnup/time
- Density varies as a function of T (thermal expansion)
- Thermal conductivity also varies with temperature

$$k_0 = \frac{100}{7.5408 + 17.629T + 3.6142T^2} + \frac{6400}{T^{5/2}} \exp\left(\frac{-16.35}{T}\right)$$



The heat capacity is a function of temperature

$$C_P = \frac{C_1 \theta^2 e^{\theta/T}}{T^2 (e^{\theta/T} - 1)^2} + 2C_2 T + \frac{C_3 E_a e^{-E_a/T}}{T^2}$$

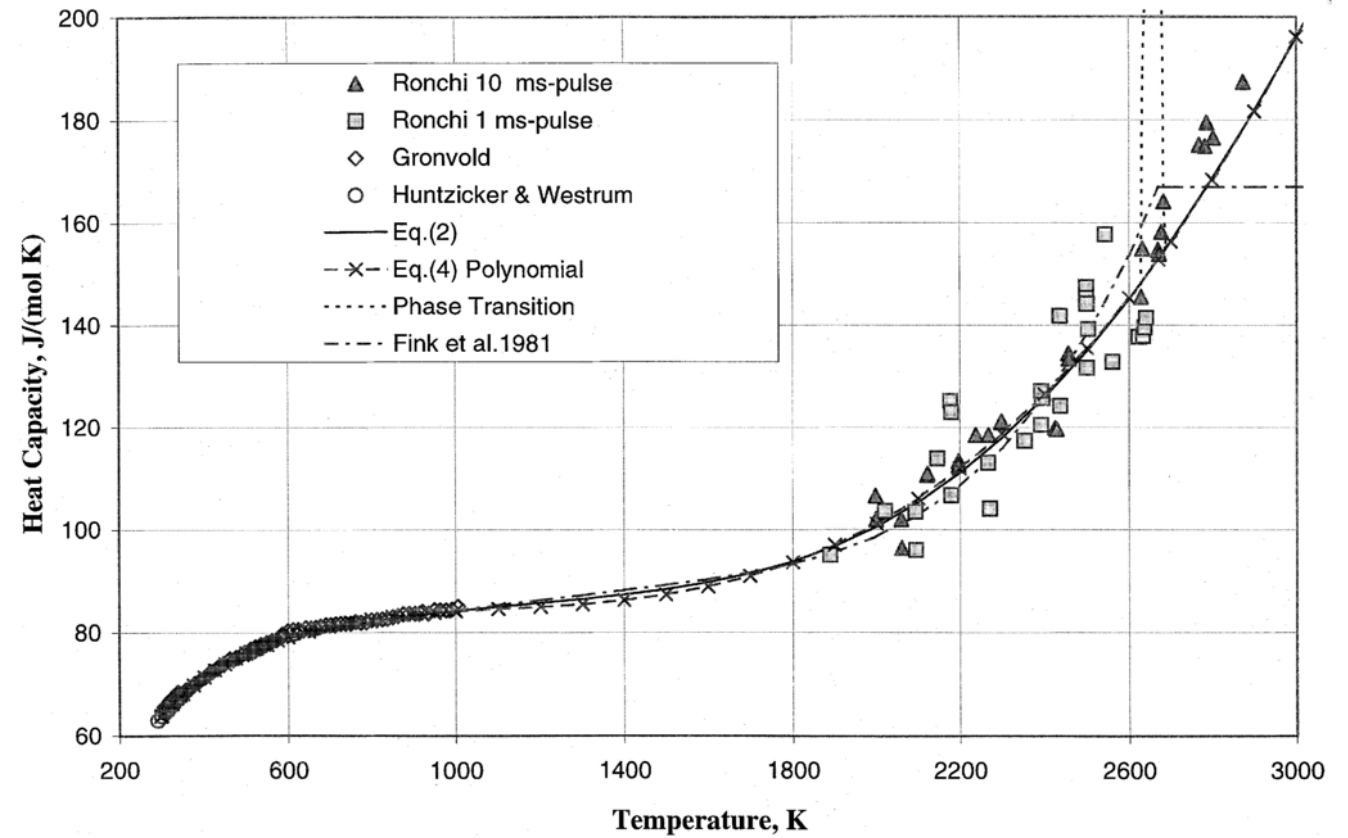
$$\theta = 548.68 \text{ K}$$

$$C_1 = 302.27 \text{ J/kg-K}$$

$$C_2 = 8.463 \times 10^{-3} \text{ J/kg-K}^2$$

$$C_3 = 8.741 \times 10^7 \text{ J/kg}$$

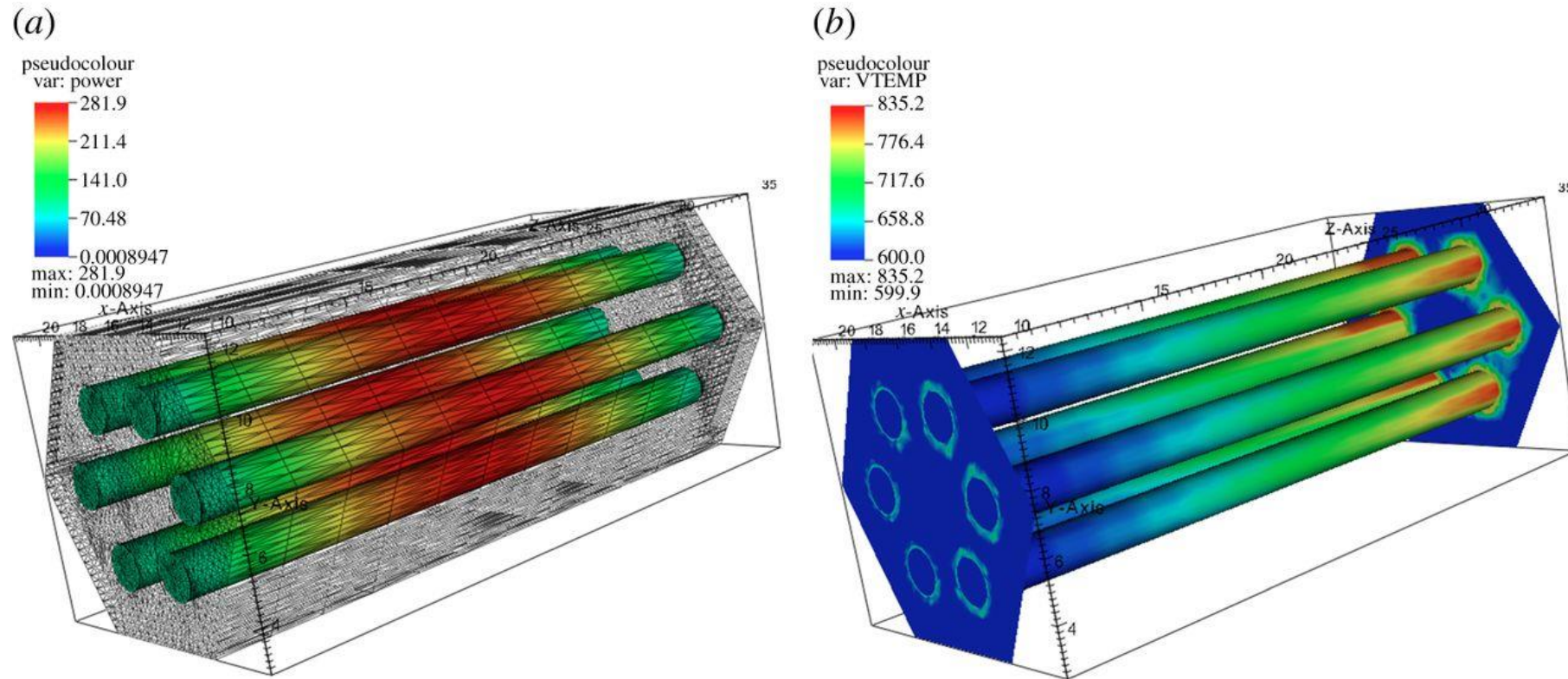
$$E_a = 18531.7 \text{ K}$$



ORNL/TM-2000/351

The heat generation rate is a function of the thermal neutron flux, which varies in time and space

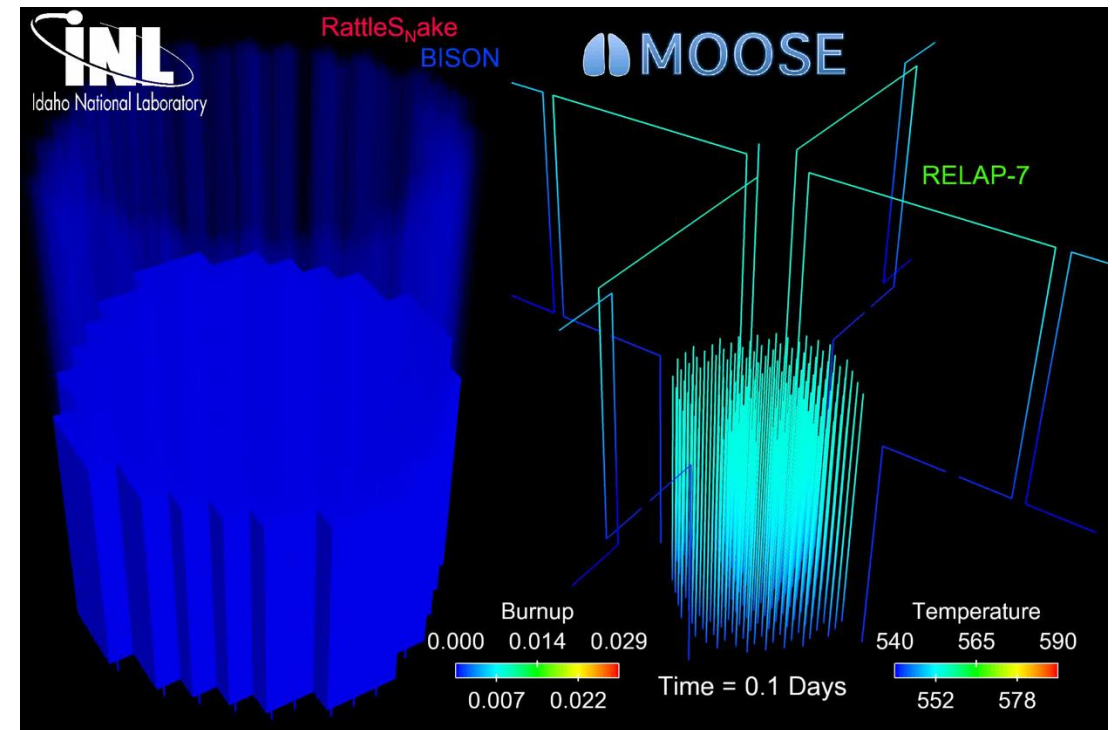
$$Q = E_f N_f \sigma_f \varphi_{th}$$



ANALYTICAL SOLVE OF HEAT CONDUCTION

The most accurate solution is numerical, in 3D, requires modeling the entire core, and is multi-physics

- Solution is 3D and changes in time
- All the properties are functions of temperature
- The boundary conditions come from information about the coolant flow
- The heat generation rate comes from information about the neutronics in the reactor
- No analytical solution is possible



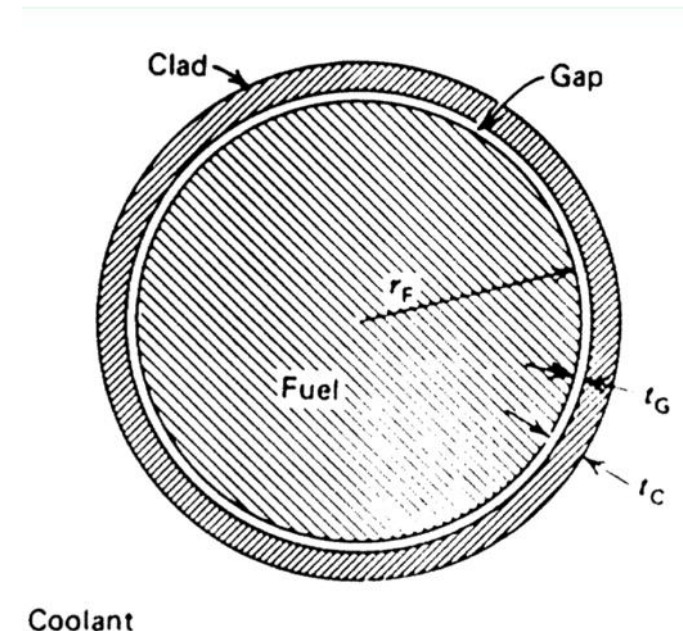
In order to solve, make assumptions!

- #1: steady state -> $\nabla \cdot (k \nabla T) + Q = 0$
- #2: cylindrical, axisymmetric ->

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r k(T) \frac{\partial T}{\partial r} \right) + \frac{\partial}{\partial z} \left(k(T) \frac{\partial T}{\partial z} \right) + Q(r, z) = 0$$
- #3: constant in z $\frac{1}{r} \frac{\partial}{\partial r} \left(r k(T) \frac{\partial T}{\partial r} \right) + Q(r) = 0$
- #4: constant thermal conductivity, volume heat

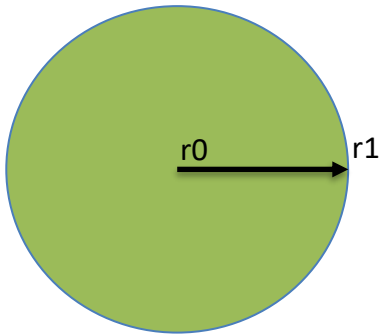
$$\frac{1}{r} \frac{\partial}{\partial r} \left(r k \frac{\partial T}{\partial r} \right) + Q = 0$$

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q$$



Directly Solving for Temperature Profile

- Boundary conditions: $r_0 = 0$, $r_1 = R$,
 $T'(0) = 0$, $T(R) = T_s$



$$\frac{1}{r} \frac{\partial}{\partial r} \left(r k \frac{\partial T}{\partial r} \right) + Q = 0$$

$$\frac{\partial}{\partial r} \left(r k \frac{\partial T}{\partial r} \right) = -Q r$$

$$r k \frac{\partial T}{\partial r} = -\frac{Q r^2}{2} + C_1 \quad 0 = -\frac{Q 0^2}{2} + C_1$$

$$\frac{\partial T}{\partial r} = -\frac{Q r}{2k}$$

$$T(r) = -\frac{Q r^2}{4k} + C_2 \quad C_2 = \frac{Q R^2}{4k} + T_s$$

$$T(r) = -\frac{Q r^2}{4k} + \frac{Q R^2}{4k} + T_s \quad T(r) = \frac{Q(R^2 - r^2)}{4k} + T_s$$

$$T_0 - T_s = \frac{Q R^2}{4k}$$

Linear Heat Rate

- $LHR = \pi R^2 Q_{av}$
 - Where Q_{av} is the radially averaged heat generation rate in W/cm³
 - LHR is in units of power per unit length: W/cm
- Substitute LHR into previous equation on $T_0 - T_s$

$$T_0 - T_s = \frac{QR^2}{4k} \qquad T_0 - T_s = \frac{R^2 LHR}{4k \pi R^2} \qquad T_0 - T_s = \frac{LHR}{4\pi k}$$

Alternate Geometries

- Similar derivation with appropriate boundary conditions can be applied to plate and sphere geometries

Plate

$$T(x) - T_s = \frac{LHR}{2\pi k} \left(1 - \frac{x^2}{t_f^2} \right)$$

x is the distance from the midplane of the fuel and t_f is the plate fuel thickness

Sphere

$$T(r) - T_s = \frac{LHR}{6\pi k} \left(1 - \frac{r^2}{R_f^2} \right)$$

r is the distance from the sphere center and R_f is the radius of the sphere

Heat transport through the gap

- Heat flux is a conserved quantity, described by Fourier's first and second laws:
- If there are no sources of heat, as if the case in the cladding and gap, the temperature field is constant with time, and the heat flux is unidirectional
- For boundary conditions T_1 and T_2 and a thickness d
- This assumes a spatially constant thermal conductivity, consistent with a "thin slab"

$$\vec{q} = -\lambda \nabla T,$$

$$\rho c \frac{\partial T}{\partial t} = \nabla(\lambda \nabla T) + q^*, \quad \lambda = \text{thermal cond.}$$

$$q = -\lambda \frac{dT(x)}{dx},$$

$$q = \frac{T_1 - T_2}{\frac{d}{\lambda}} \quad T(x) = T_1 - \frac{T_1 - T_2}{d} x$$

Heat transport through the gap

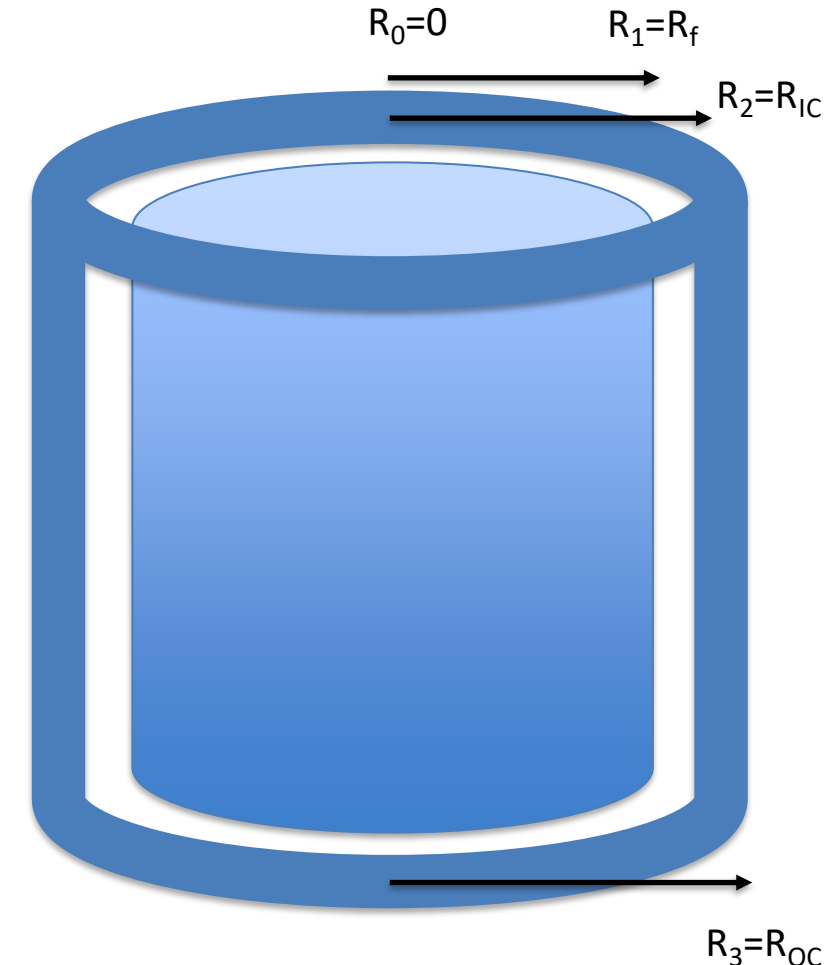
- The heat flux gives the rate, per unit area, at which heat flows in a given direction

$$q = -k \frac{T_2 - T_1}{d} \quad q_{gap} = -k_{gap} \frac{T_{IC} - T_{fuel}}{R_{IC} - R_{fuel}}$$

- The heat flux from the fuel is the LHR/pellet circumference

$$q = \frac{LHR}{2\pi R_f}$$

- Heat flux from the fuel is the same as heat flux through the the gap – can make this assumption because $R_f \gg t_g$
- Gap thickness = $R_{IC} - R_f = t_g$
- Cladding thickness = $R_{OC} - R_{IC} = t_c$



Heat transport through the gap

- Set heat flux fuel/gap equal

$$\frac{LHR}{2\pi R_f} = -k_{gap} \frac{T_{IC} - T_{fuel}}{t_{gap}} \quad \frac{LHR}{2\pi R_f} = k_{gap} \frac{T_{fuel} - T_{IC}}{t_{gap}}$$

- Gap conductance is defined as:

$$h_g = \frac{k_{gap}}{t_g} \quad T_{fuel} - T_{IC} = \frac{LHR}{2\pi R_f h_g}$$

- Gap conductance depends on the gas filling the gap
 - For pure He, $k_{gap} = 16 \times 10^{-6} * T^{0.79}$ (W/cm-K)
 - For pure Xe, $k_{gap} = 0.7 \times 10^{-6} * T^{0.79}$ (W/cm-K)
 - Simple mixing rule: $k_{gap} = k_{He}(1-y) + k_{Xe}y$
 - Where y is the mole/atom fraction of Xe

Heat transport through the cladding

- Heat flux through the cladding $q = -k \frac{T_2 - T_1}{d}$ $q_{clad} = -k_{clad} \frac{T_{OC} - T_{IC}}{R_{OC} - R_{IC}}$

$$q = \frac{LHR}{2\pi R_f} \quad q_{clad} = k_{clad} \frac{T_{IC} - T_{OC}}{t_{clad}}$$

- Set equal your heat fluxes

$$\frac{LHR}{2\pi R_f} = k_{clad} \frac{T_{IC} - T_{OC}}{t_{clad}} \quad T_{IC} - T_{OC} = \frac{LHR}{2\pi R_f} \frac{t_{clad}}{k_{clad}}$$

Heat transfer to the coolant

- Heat is transported from the cladding to the coolant via convection

$$T_{CO} - T_{cool} = \frac{LHR}{2\pi R_F h_{cool}}$$

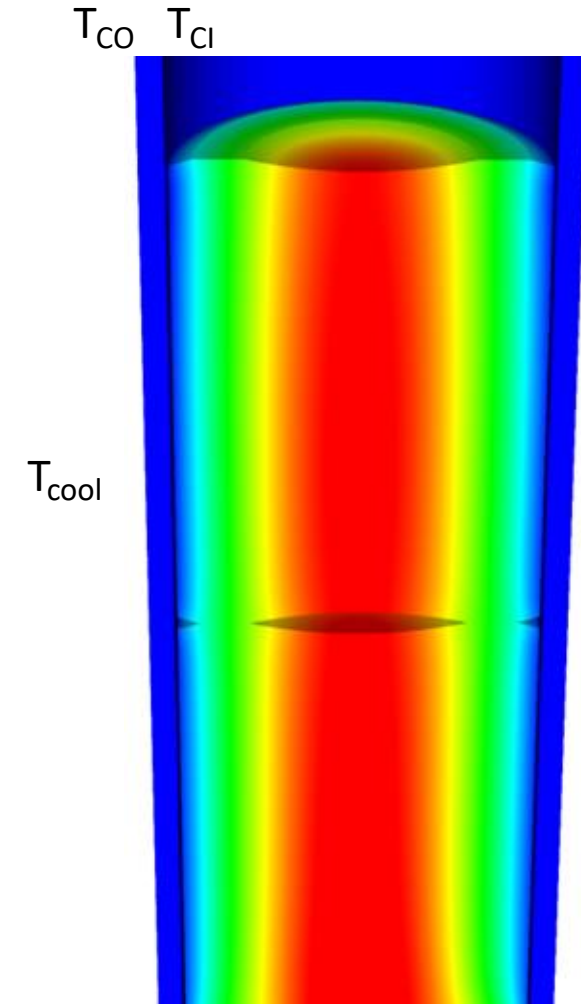
- T_{cool} is the coolant temperature, h_{cool} is the convective heat transfer coefficient between cladding wall and coolant
- Adding gap + cladding + coolant: $\frac{1}{h} = \frac{t_{gap}}{k_{gap}} + \frac{t_{clad}}{k_{clad}} + \frac{1}{h_{cool}}$

Summary of analytical solutions

- $T_0 - T_{fuel} = \frac{Q}{4k} R_{fuel}^2 \quad T_0 - T_{fuel} = \frac{LHR}{4\pi k}$
- $T_{fuel} - T_{IC} = \frac{Q}{2h_{gap}} R_{fuel} \quad T_{fuel} - T_{IC} = \frac{LHR}{2\pi R_{fuel} h_{gap}} \quad h_{gap} = \frac{k_{gap}}{t_{gap}}$
- $T_{IC} - T_{OC} = \frac{Q t_{clad}}{2k_{clad}} R_{fuel} \quad T_{IC} - T_{OC} = \frac{LHR t_{clad}}{2\pi R_{fuel} k_{clad}}$
- $T_{OC} - T_{cool} = \frac{Q}{2h_{cool}} R_{fuel} \quad T_{OC} - T_{cool} = \frac{LHR}{2\pi R_{fuel} h_{cool}}$

Solving for the temperature profile

- You solve for the transition temperatures
- Start from the coolant (known BC) and work inward
- Have a linear profile everywhere except in the fuel
- Finally, solve for the temperature profile throughout the fuel



Fuel and Cladding Thermal Properties

Material	Density (g/cm ³)	Heat Capacity C _p (J/g-K)	Thermal Conductivity k (W/cm-K)	Thermal Expansion Coefficient α (K ⁻¹)
UO ₂	10.98	0.33	0.03	1.2 x 10 ⁻⁵
Zr	6.5	0.35	0.17	1.0 x 10 ⁻⁵
Stainless steel	8.0	0.5	0.17	9.6 x 10 ⁻⁶

Example Problem

Temperature profile calculation example

- $T_{cool} = 580 \text{ K}$; $LHR = 200 \text{ W/cm}$; $h_{cool} = 2.65 \text{ W/cm}^2\text{-K}$
- $R_{fuel} = 0.5 \text{ cm}$; $t_{clad} = 0.06 \text{ cm}$; $t_{gap} = 0.003 \text{ cm}$; $k_f = 0.03 \text{ W/cm-K}$
- Work from outside->in, calculate cladding temperature

$$T_{co} = (200)/(2 \cdot \pi \cdot 0.5 \cdot 2.65) + 580$$

$$T_{co} - T_{cool} = \frac{LHR}{2\pi R_{fuel} h_{cool}}$$

$$T_{co} = 604 \text{ K}$$

- Calculate inner cladding temp

$$T_{ci} = (200 \cdot 0.06)/(2 \cdot \pi \cdot 0.5 \cdot 0.17) + 604$$

$$T_{ci} - T_{clad} = \frac{LHR t_{clad}}{2\pi R_{fuel} k_{clad}}$$

$$T_{ci} = 626 \text{ K}$$

Temperature profile calculation example

- Calculate fuel surface temperature
- Calculate gap conductance
 - gap with He; $k_{\text{gap}} = 16 \times 10^{-6} * T^{0.79}$ (W/cm-k); assume T_{ci} is appropriate for entire gap; $k_{\text{gap}} = 0.0026$ W/cm-K; $t_{\text{gap}} = 0.003$ cm
 - $h_{\text{gap}} = 0.87$ W/cm²-K

$$T_{\text{fuel}} = 200 / (2 * \pi * 0.5 * 0.87) + 626.5$$

$$T_{\text{fuel}} - T_{ci} = \frac{LHR}{2\pi R_{\text{fuel}} h_{\text{gap}}}$$

$$T_{\text{fuel}} = 700.3 \text{ K}$$

Temperature profile calculation example

- Calculate centerline temperature

$$T_0 = 200/(4*\pi*0.03) + 700.3$$

$$T_0 - T_{fuel} = \frac{LHR}{4\pi k}$$

$$T_0 = 1230.8 \text{ K}$$

Full temperature profile

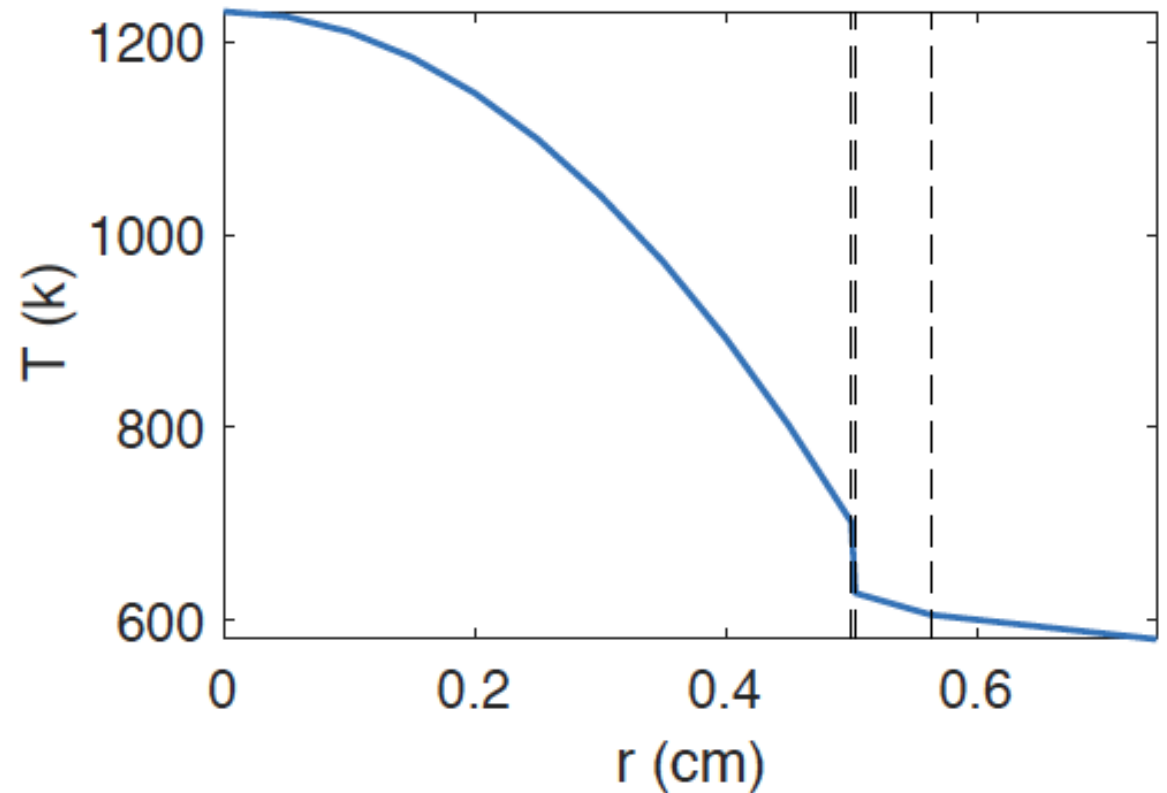
$$T(r) = \frac{Q(R^2 - r^2)}{4k} + T_s$$

$$T(r) = \frac{LHR(R_f^2 - r^2)}{4\pi k R_f} + T_s$$

$$T(r) = \frac{LHR}{4\pi k} \left(1 - \frac{r^2}{R_f^2} \right) + T_s$$

Temperature profile calculation example

- Parabolic temperature profile in fuel
- Linear in gap and clad
- Steep temperature drop over gap, very low thermal conductivity
- Smaller temperature drop over cladding

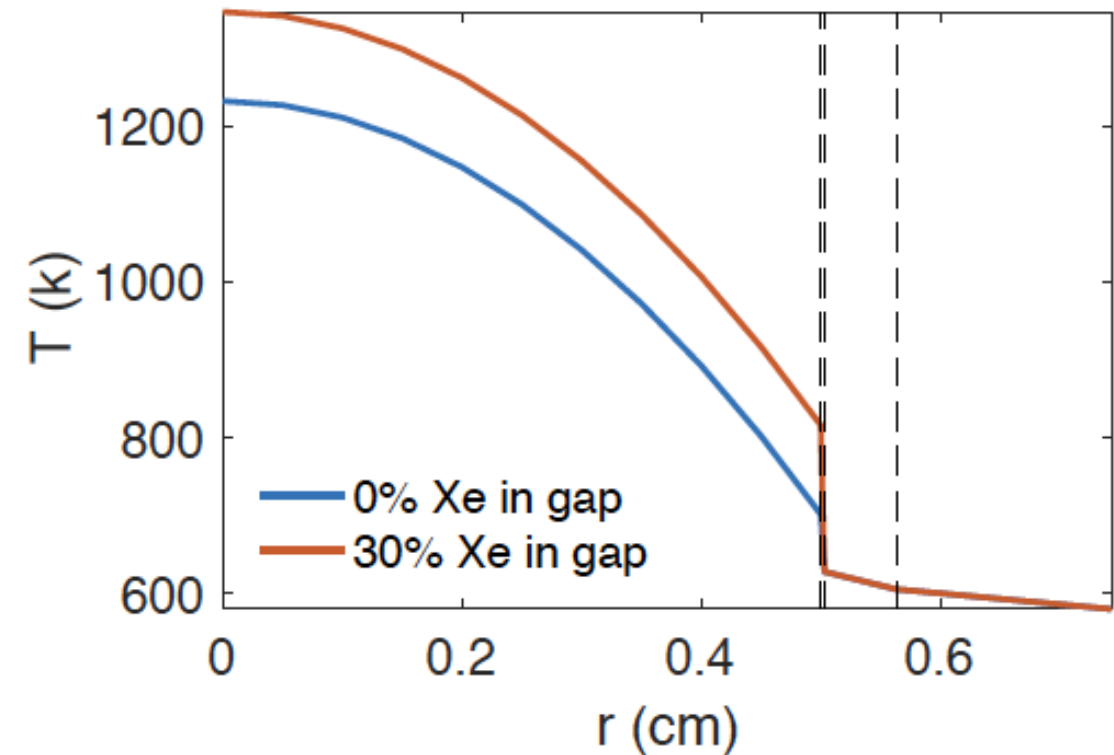


Temperature profile modification

- Instead of pure He gap, 30% of gas is Xe; how is T_0 affected?
- First, how is gap conductance affected?
- Gap conductance depends on the gas filling the gap
 - For pure He, $k_{\text{gap}} = 16 \times 10^{-6} * T^{0.79}$ (W/cm-K)
 - For pure Xe, $k_{\text{gap}} = 0.7 \times 10^{-6} * T^{0.79}$ (W/cm-K)
 - Simple mixing rule: $k_{\text{gap}} = k_{\text{He}}(1-y) + k_{\text{Xe}}y$
 - T_{Cl} , T_{CO} , and T_{cool} is unchanged from previous example, also $T_0 - T_{\text{fuel}}$ is unchanged
 - $k_{\text{gap}} = ((16 \times 10^{-6}) * (626)^{0.79})(1-0.3) + ((0.7 \times 10^{-6}) * (626)^{0.79})(0.3) = 1.85 \text{E-3 W/cm-K}$

Temperature profile modification

- $k_{\text{gap}} = 1.85\text{E-}3 \text{ W/cm-K}$
- $h_{\text{gap}} = 1.85\text{E-}3 / 0.003 = 0.62 \text{ W/cm}^2\text{-K}$
- $T_{\text{fuel}} = 200 / (2 \cdot \pi \cdot 0.5 \cdot 0.62) + 626 = 729 \text{ K}$
- $T_0 - T_{\text{fuel}} = 530.5 \text{ K}$ (unchanged from before)
- $T_0 = 729 + 530.5 = 1259.5 \text{ K}$
- Increase in T_0 of 30 K
- Caveat: linear mixing of gases is not the best approach



Summary

- Developed analytical expressions for the temperature profile within a fuel rod
- Required to make four assumptions:
 - Steady-state solution
 - Temperature is axisymmetric
 - T is constant in Z
 - Thermal conductivity is independent of temperature
- Temperature profile in the fuel is parabolic, assume linear profiles in gap, clad and coolant
- Can incorporate axial variation in T_{cool} with axial variation in LHR